

On the Telluric Electric Frequency Selection Method (TEFSM) for geophysical groundwater exploration

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THROUGH CONJUNCTIVE WATER MANAGEMENT

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Introduction

- ❖ The success of groundwater supply projects largely depends on the ability to locate feasible groundwater sources.
- ❖ Many approaches exist for prospecting groundwater, but the geophysical approach remains the conventional approach.
- ❖ However, the high cost of traditional geophysical approaches remains a hindrance, particularly in low-income regions such as Africa.



Introduction

- ❖ The 20th century has witnessed the emergence of new geophysical approaches to groundwater exploration.
- ❖ The Telluric Electric Frequency Selection Method (TEFM) is one recent technology.
- ❖ The TEFM uses natural electromagnetic (EM) waves to evaluate the groundwater potential of the subsurface rocks.
- ❖ Reported studies to shows that the approach has great potential (Isah et al. 2022; Agyemang 2022; Song et al. 2021; Yulong et al. 2023; Gomo 2023; Lu et al. 2023; Gomo 2024a)



ADMT 400 SX 32-D: (Shanghai Aidu Energy Technology Co Ltd, 2022)



PQWT-GT500A: <https://www.pqwtdetector.com/>

Introduction

- ❖ However, several issues are emerging about the TEFSM method, and these warrant a discussion to provide an understanding of the capabilities and limitations of the TEFSM geophysical approach based on field data and reported studies.
- ❖ This presentation provides an overview of the principles of the TEFSM according to the available literature.
- ❖ Share field experiences on the strengths and limitations of the approach.
- ❖ Recommendations for future studies to optimise the applications of the TEFSM approach are made



ADMT 400 SX 32-D: (Shanghai Aidu Energy Technology Co Ltd, 2022)



PQWT-GT500A: <https://www.pqwtdetector.com/>

Basic principles of the TEFSM

- Secondary EM fields are induced within the Earth by the time-varying fields in such a way that the generation of secondary fields is dependent on the electrical conductivity or resistivity of the Earth.
- However, in the natural electric field approach the generation of secondary fields is rather controlled by the shielding effectiveness.
- The shielding effectiveness is defined as the ability of subsurface geological materials to absorb the induced EM waves (Hunan Puqi Geologic Exploration Equipment Institute, 2017).
- In the uniform earth, the impedance is related to the frequency and resistivity of the earth and the electromagnetic field to evaluate the earth's apparent resistivity using:

$$\rho = \frac{1}{5f} \left| \frac{E}{H} \right|^2$$

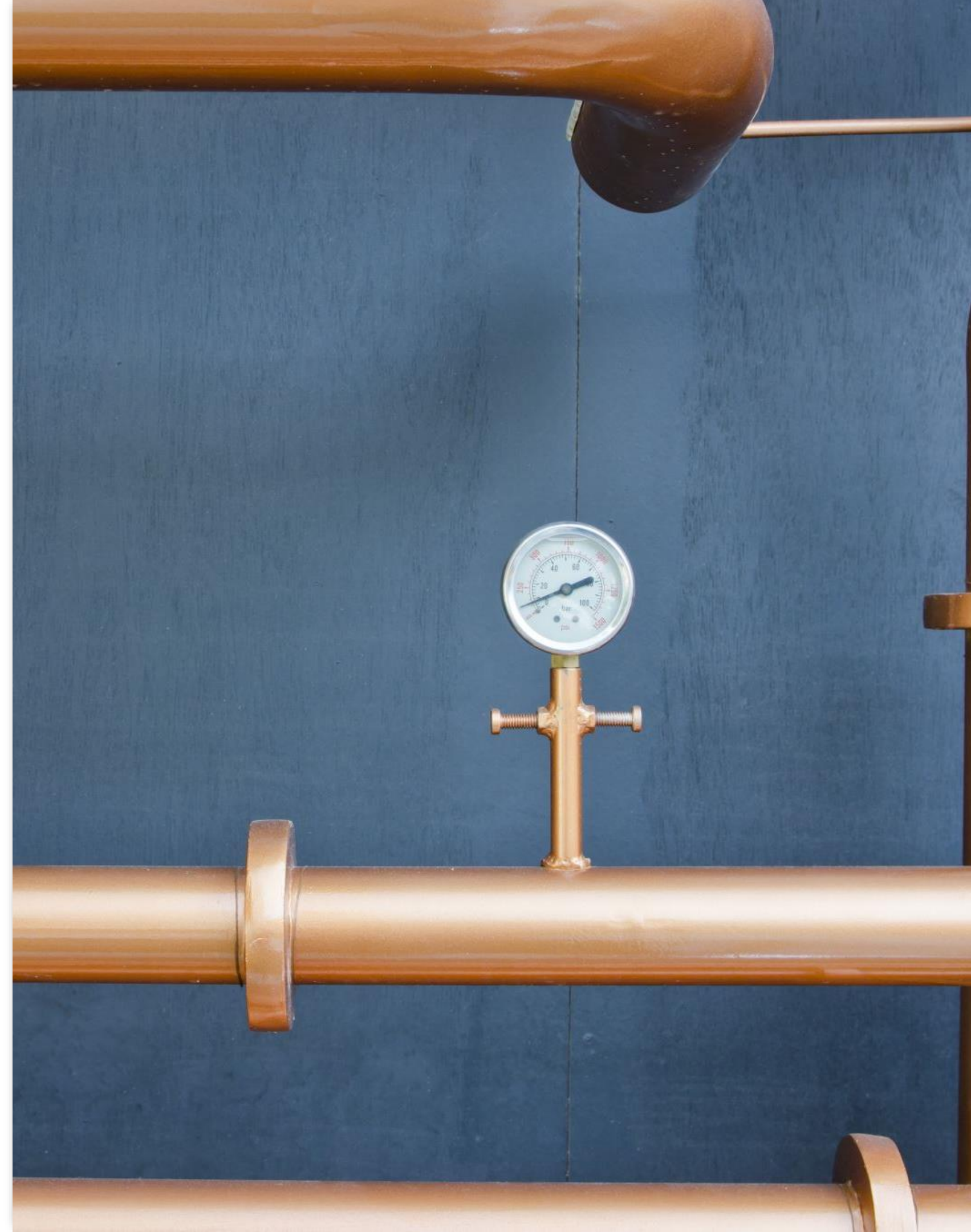
- Where: E is the electric field component, H is the magnetic field component, f is the frequency of the EM waves
- In a smaller area, the magnetic field can be considered stable and therefore assumed to be constant, such that the *qualitative relationship between the electric field component and resistivity* is used to assess the electrical properties of the geological bodies.

TEFSM Basic principles

- The investigation depth (δ) of the propagating waves according to the attenuation properties of the plane is evaluated using the frequency and resistivity in Equation:

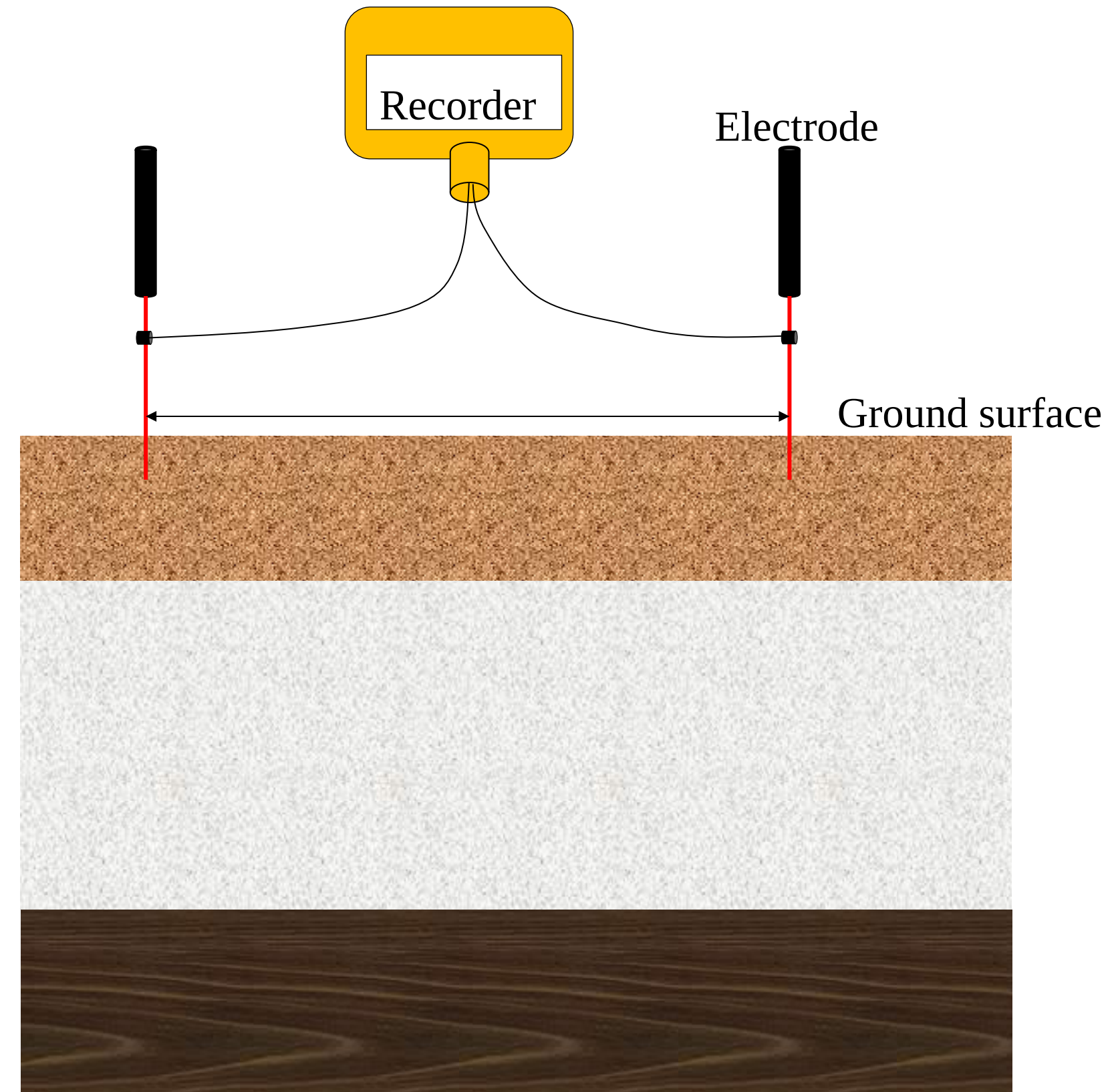
$$\delta \approx 503 \sqrt{\frac{\rho}{f}}$$

- When the resistivity of the underground materials is assumed to have a negligible effect, the investigation depth is determined only from the wave frequency (Hunan Puqi Geologic Exploration Equipment Institute, 2017)
- The different frequencies of the electric field strength reflect the earth's electrical structure at different depths



TEFSM Basic principles

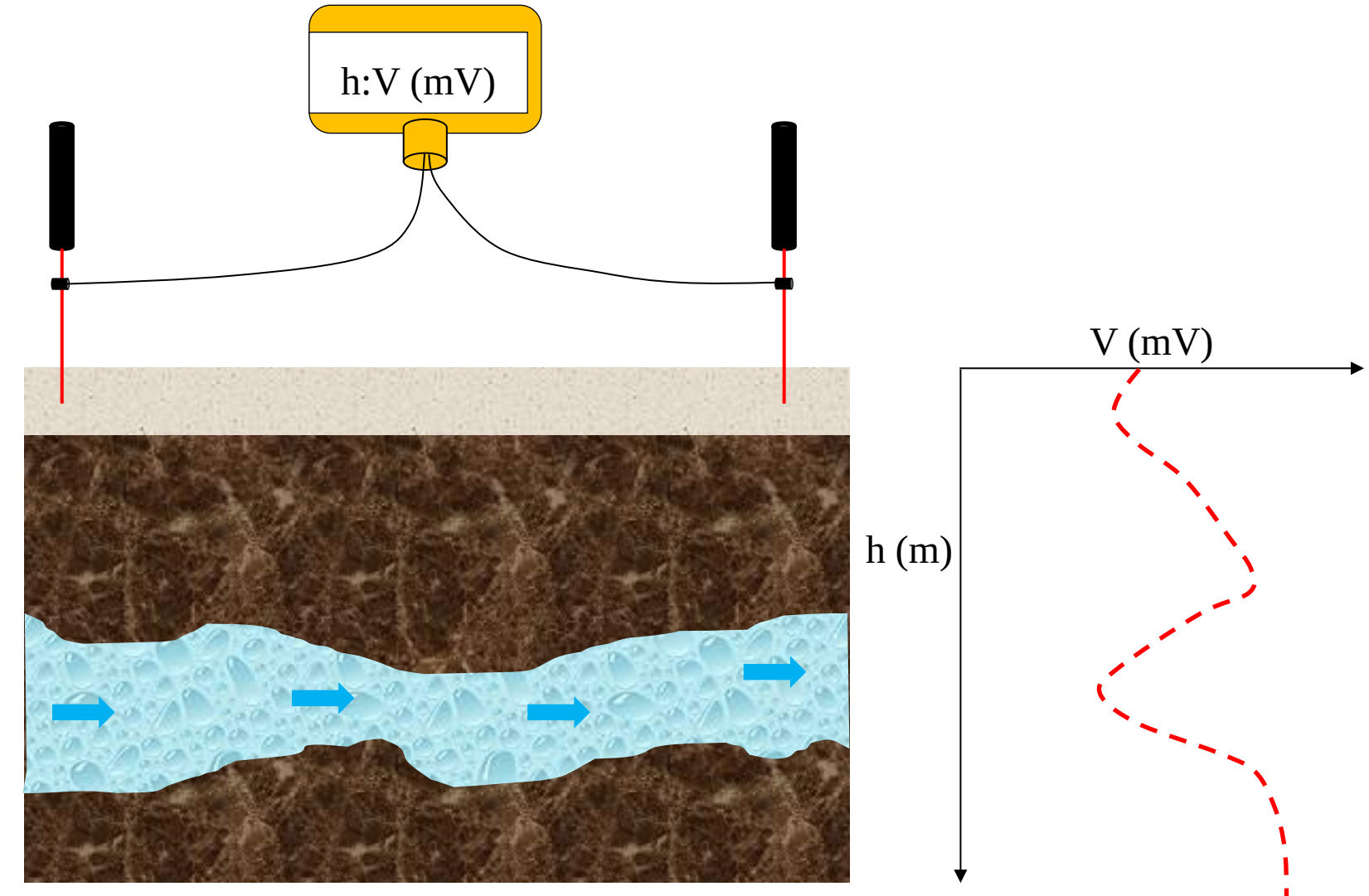
- Signals of the electric field of different frequencies and therefore at different depths are measured as the electrical potential difference between two grounded copper electrodes
- The electric potential difference output readings reflect the shielding effectiveness of earth's subsurface materials to electric field waves at certain depths and are used as the investigation parameter.
- It should therefore be clear that this approach does not evaluate the subsurface resistivity as has been traditionally applied (PQWT 2022).



TEFSM Basic principles

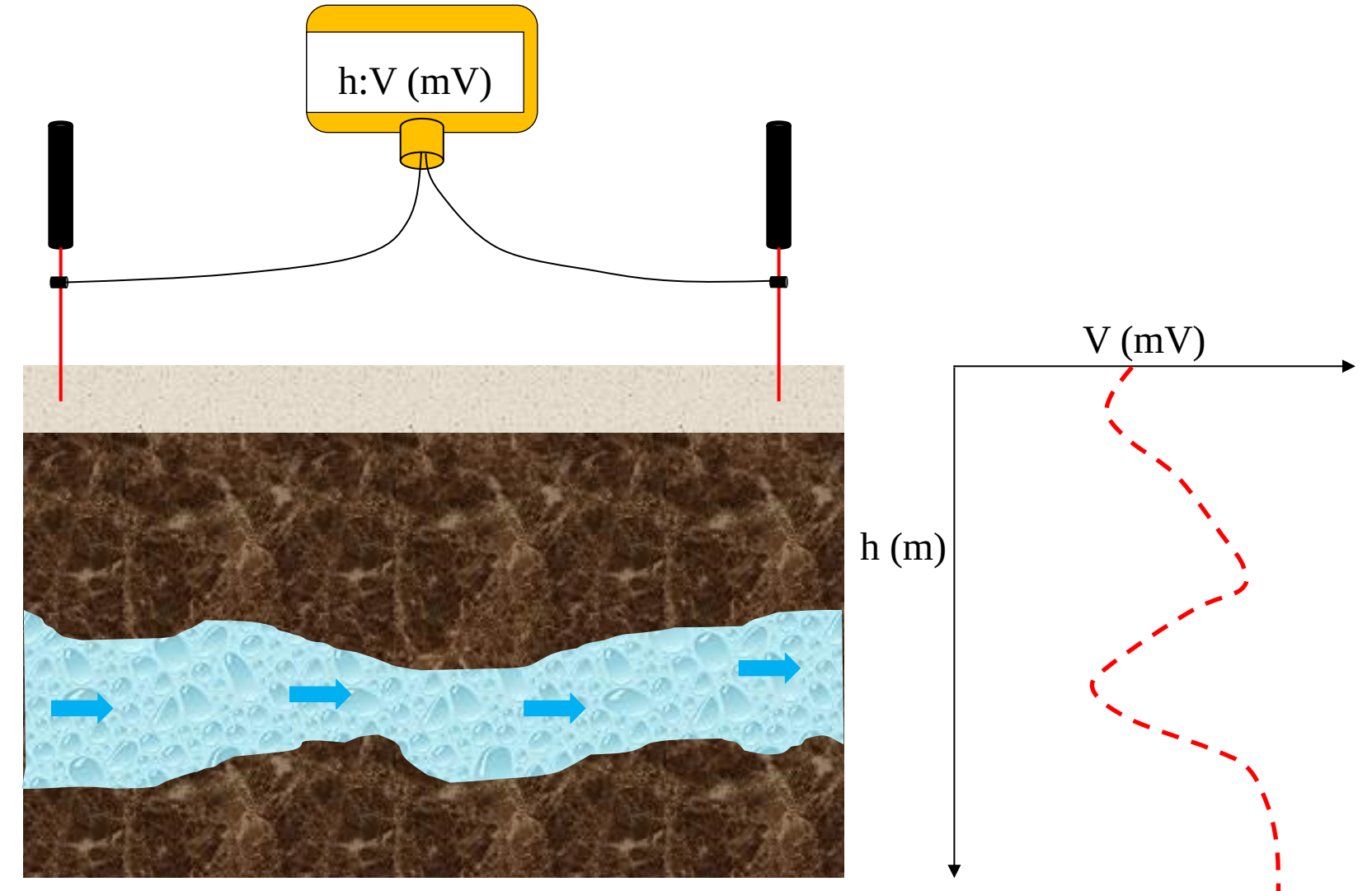
Use of electrical potential difference

- Due to the composition of polar molecules water has greater capacity to absorb the propagating induced EM waves and hence high shielding effectiveness (Hunan Puqi Geologic Exploration Equipment Institute, 2017).
- The high shielding effectiveness result in low EPD difference being measured at the surface.
- The same can be said for weathered and or fractured zones of the hard rock aquifers due to their potential to store and transmit groundwater.
- The TEFSM equipment is commonly known as **Groundwater Detectors**



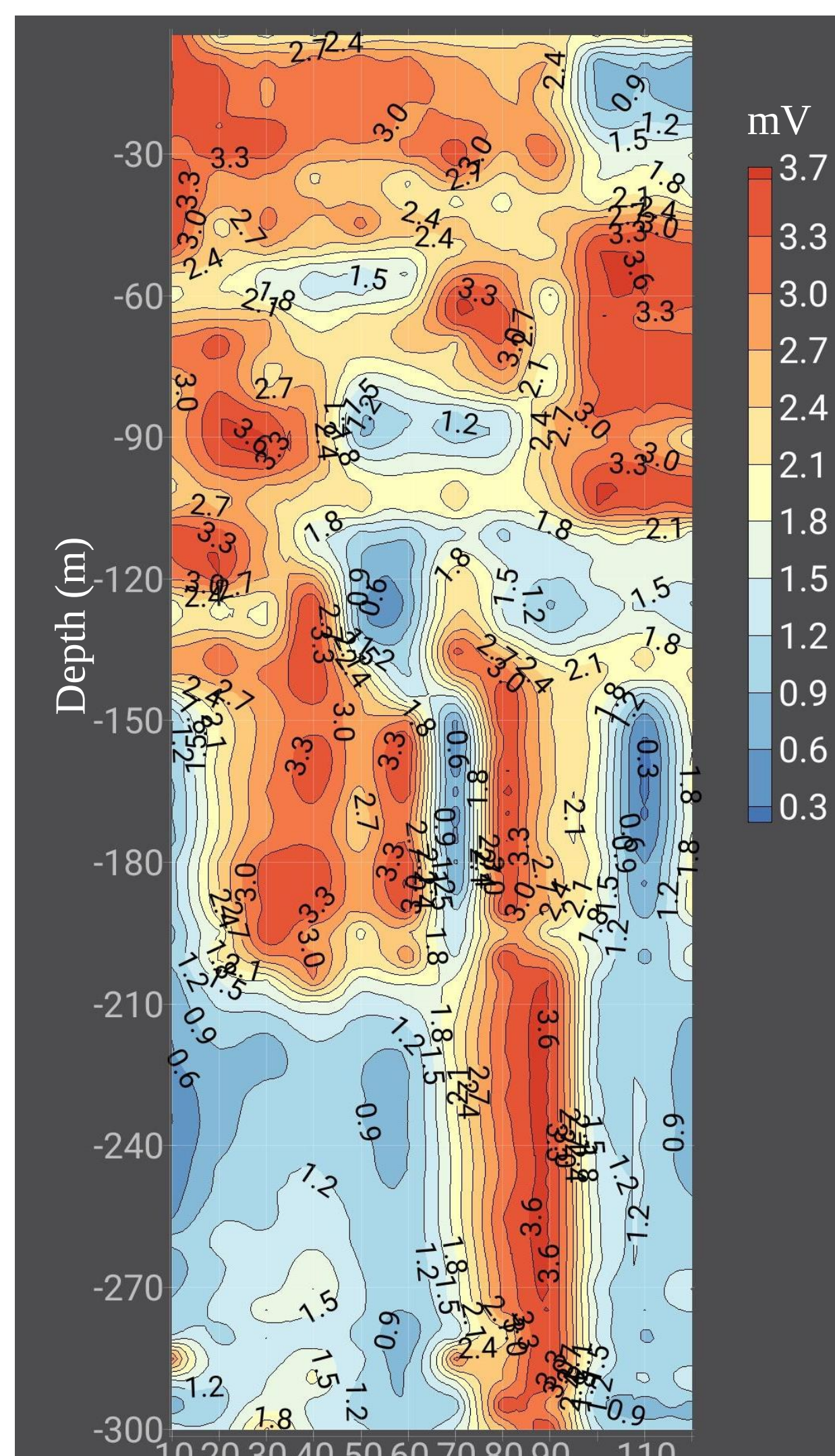
TEFSM field data collection

- Electric field measurements at different frequencies corresponding to different depths (h) are recorded as the electrical potential difference (V) between two grounded copper electrodes.
- The station of measurement is located at the centre of the two-electrodes. The basic TEFSM equipment only makes one measurement at a station and time.
- Electrodes are moved along the survey line to make measurements at subsequent stations.
- The advanced TEFSM technology has multiple electrodes.



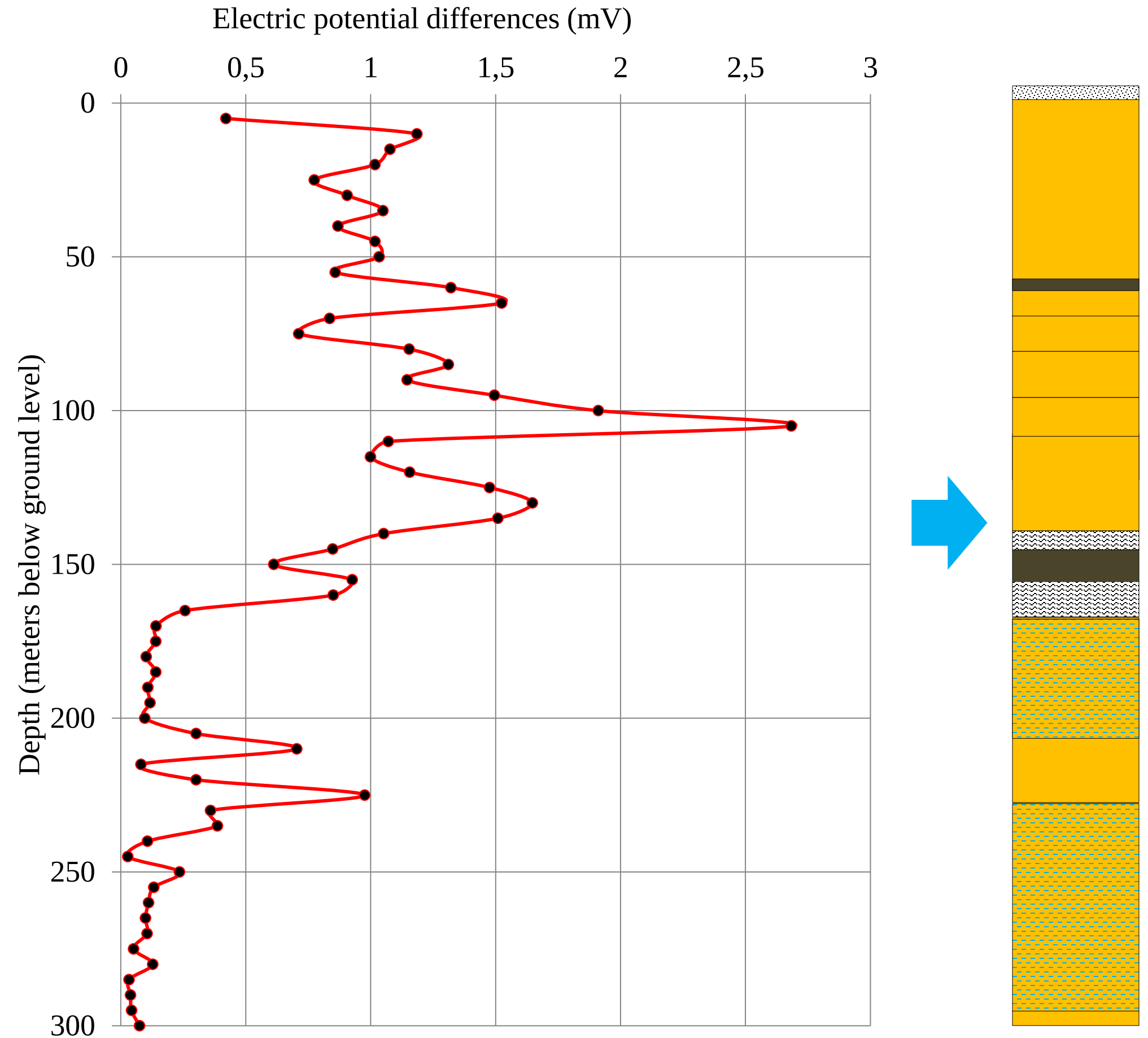
TEFSM data processing and analysis

- There are different approaches to processing and analysing TEFSM data.
- The first approach uses an inbuilt software in the TEFSM meter and can create a 2-D pseudo section of electric potential difference along the traverse line (Hunan Puqi Geologic Exploration Equipment Institute 2017b; Aidu Energy Technology 2020).
- Anomalies can be related to underlying geological and or hydrogeological properties to identify groundwater potential.
- Aidu Energy Technology (2020) also provides an option to send data for analysis by their experts.



TEFSM data processing and analysis

- The other approach involves downloading the data, plotting the electric potential difference against depth.
- Changes in electrical potential differences with time are correlating to drilling data or the general hydrogeology of the area from literature to delineate potential aquifers.



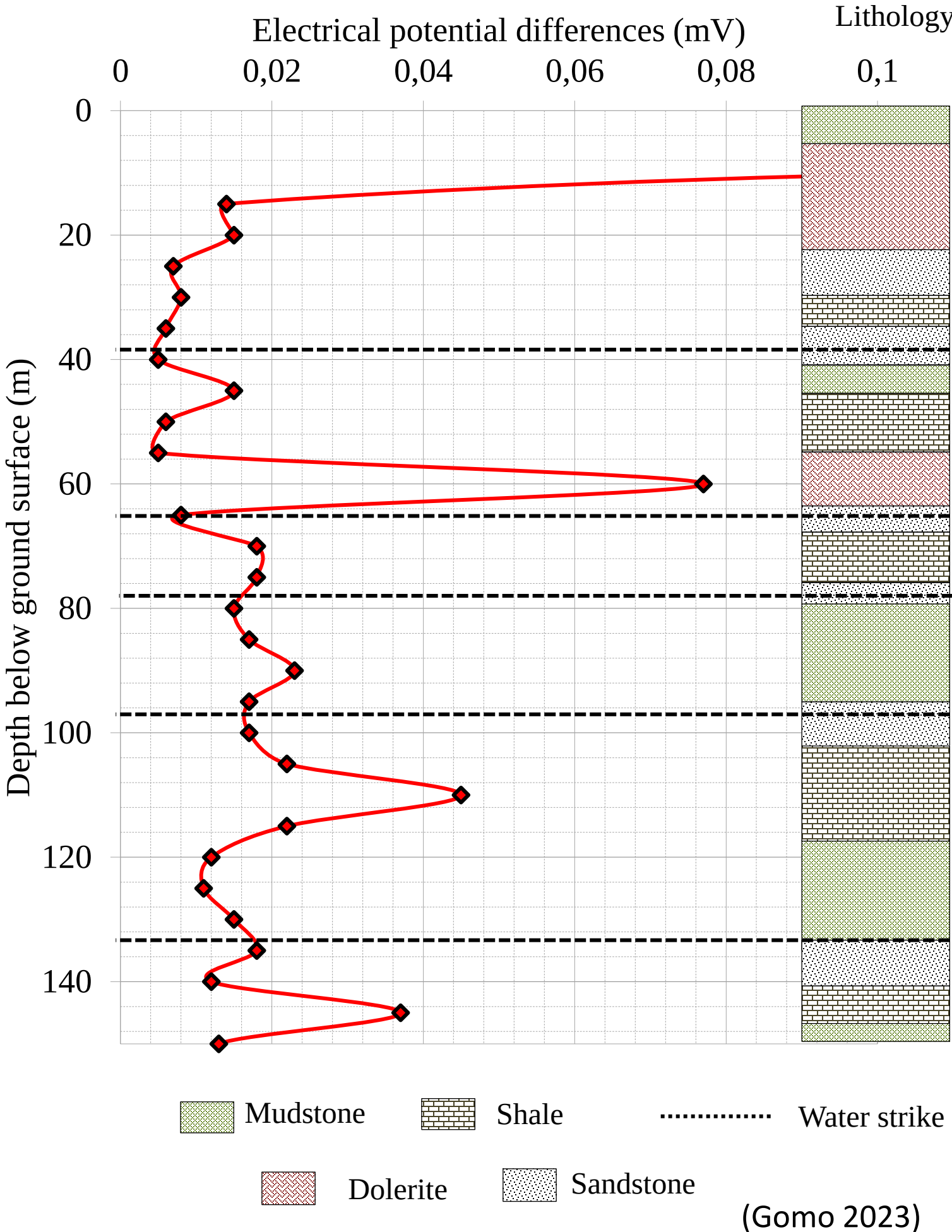
Some examples of the TEFSM application in groundwater exploration

Our field experience

Karoo sandstone aquifers in the Free State Province, South Africa (Gomo 2023)

TEFSM measurements corroborated with drilling data (> 10 boreholes)

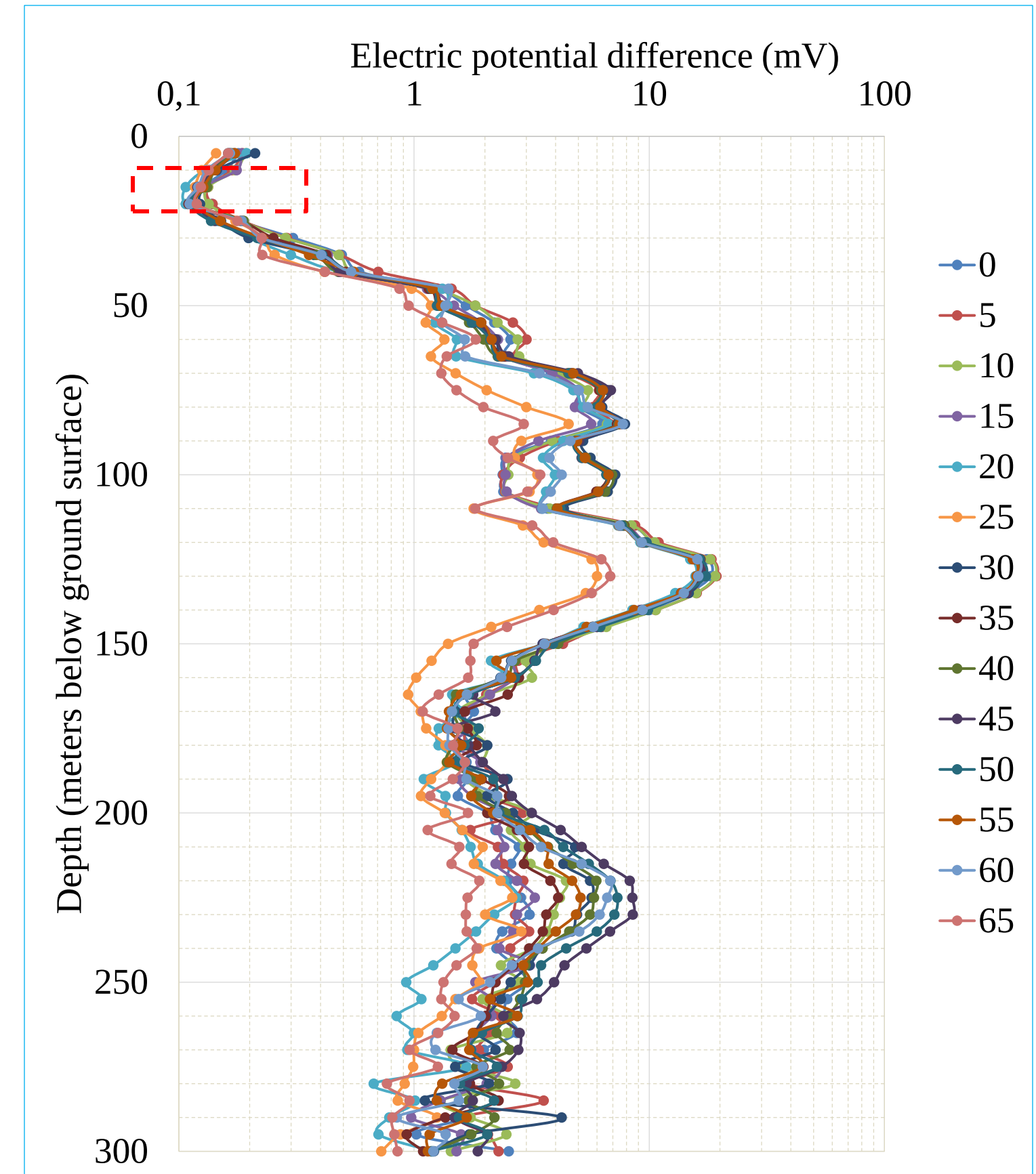
Multiple layers of sandstone aquifers detected up 150 m depth, EPD range: 0.005 to 0.075 mV.



Some examples of the TEFSM application in groundwater exploration

Ongoing research

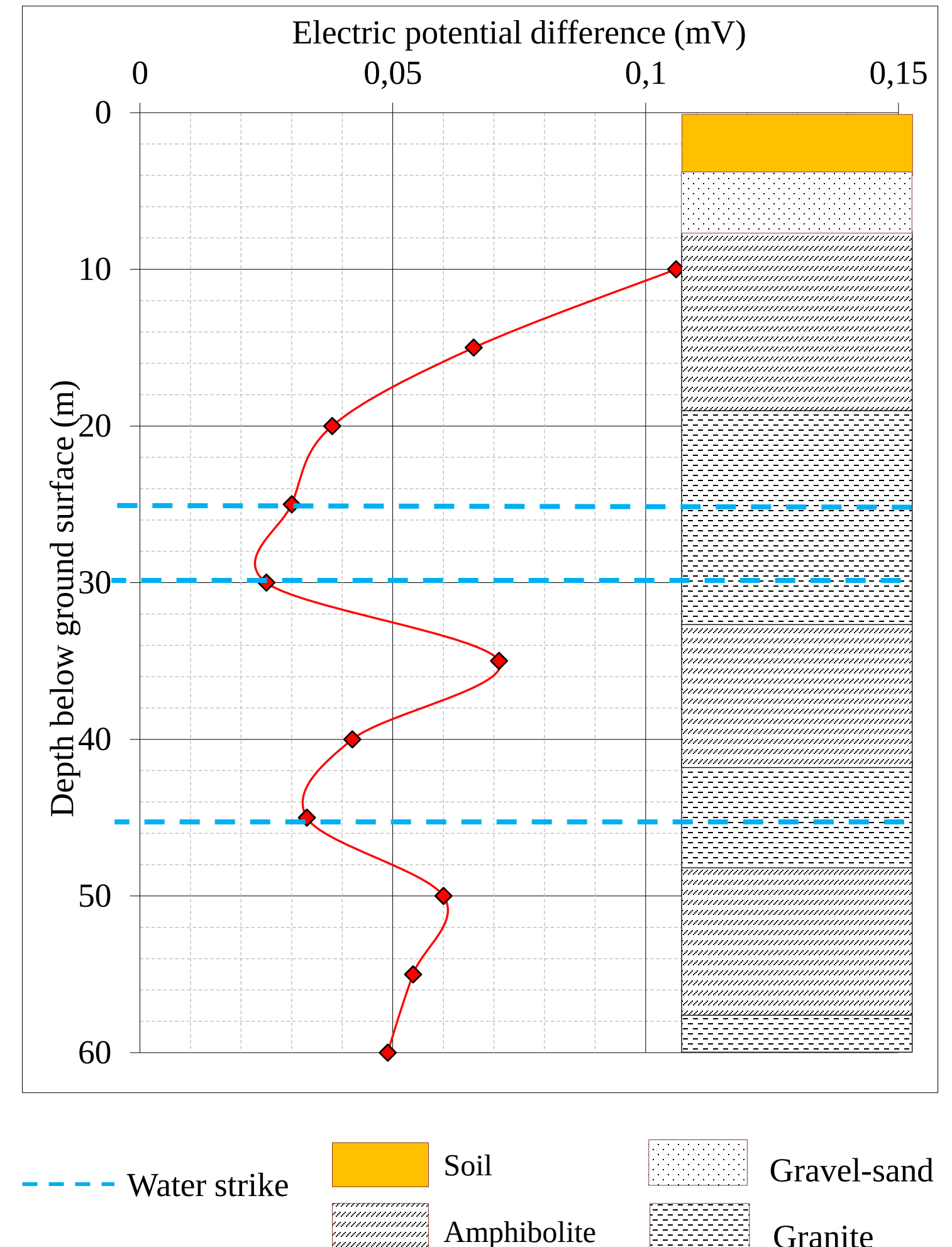
- Example of TEFSM data from University of the Free State campus test site
- Sandstone aquifer located ~ 15 to 25 mbgs; bedding plane fracture ~ 20 mbgs)
- The TEFSM shows clear distinctive anomalies at these depth



Examples of TEFSM application in groundwater exploration

Our field experience

- Weathered granite aquifers in Eswatini and Zimbabwe (Gomo and Ngobe, 2024)
- TEFSM corroborated with drilling data (> 30 boreholes)
- Multiple layers of weathered granite aquifers delineated up to 120 m depth, EPD range: 0.018-0.068 mV



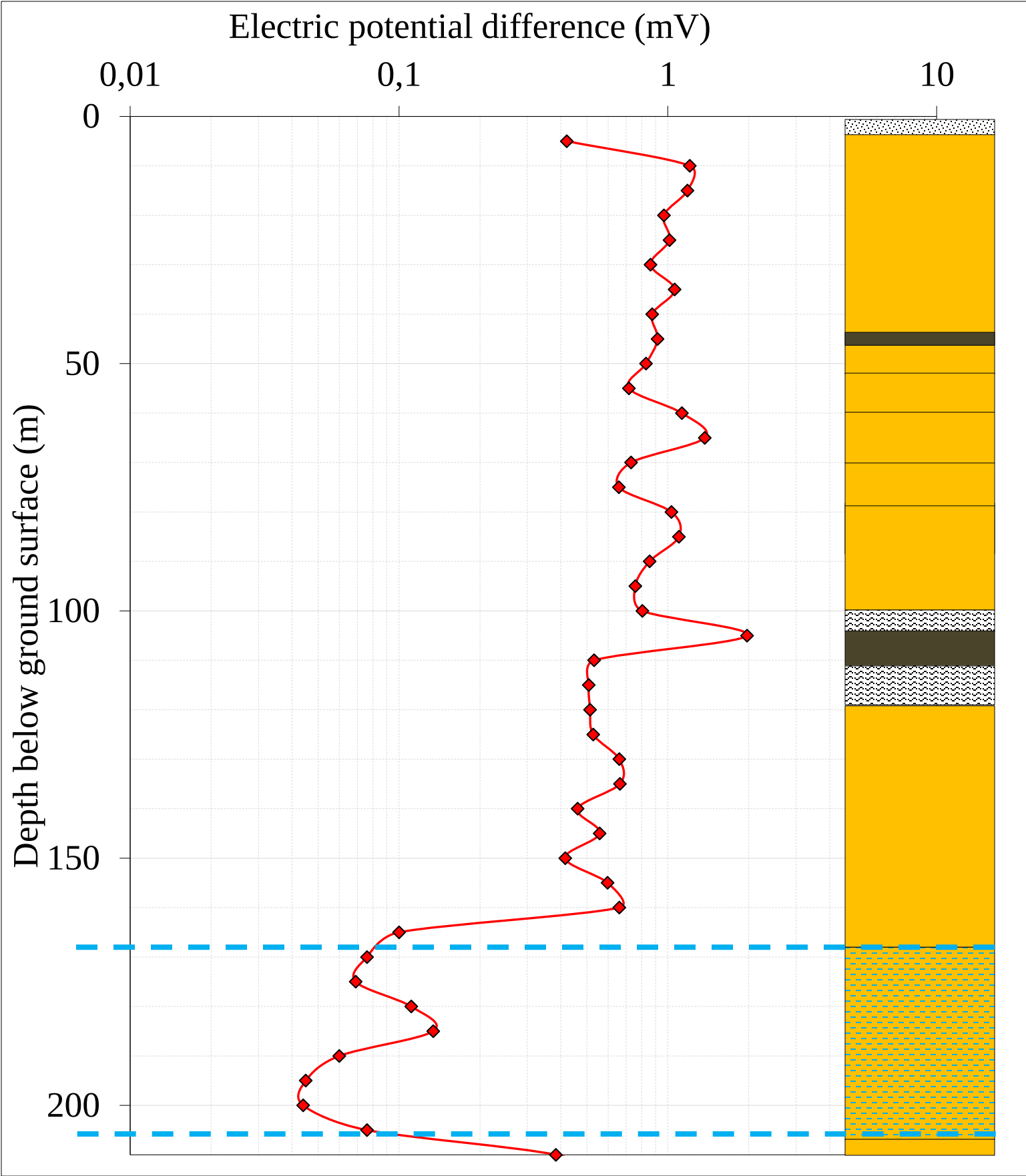
Examples of TEFSM application in groundwater exploration

Our field experience

Deeper exploration of dolomite aquifer in Postmasburg, South Africa (Gomo 2024)

TEFSM corroborated with drilling data (2 Boreholes)

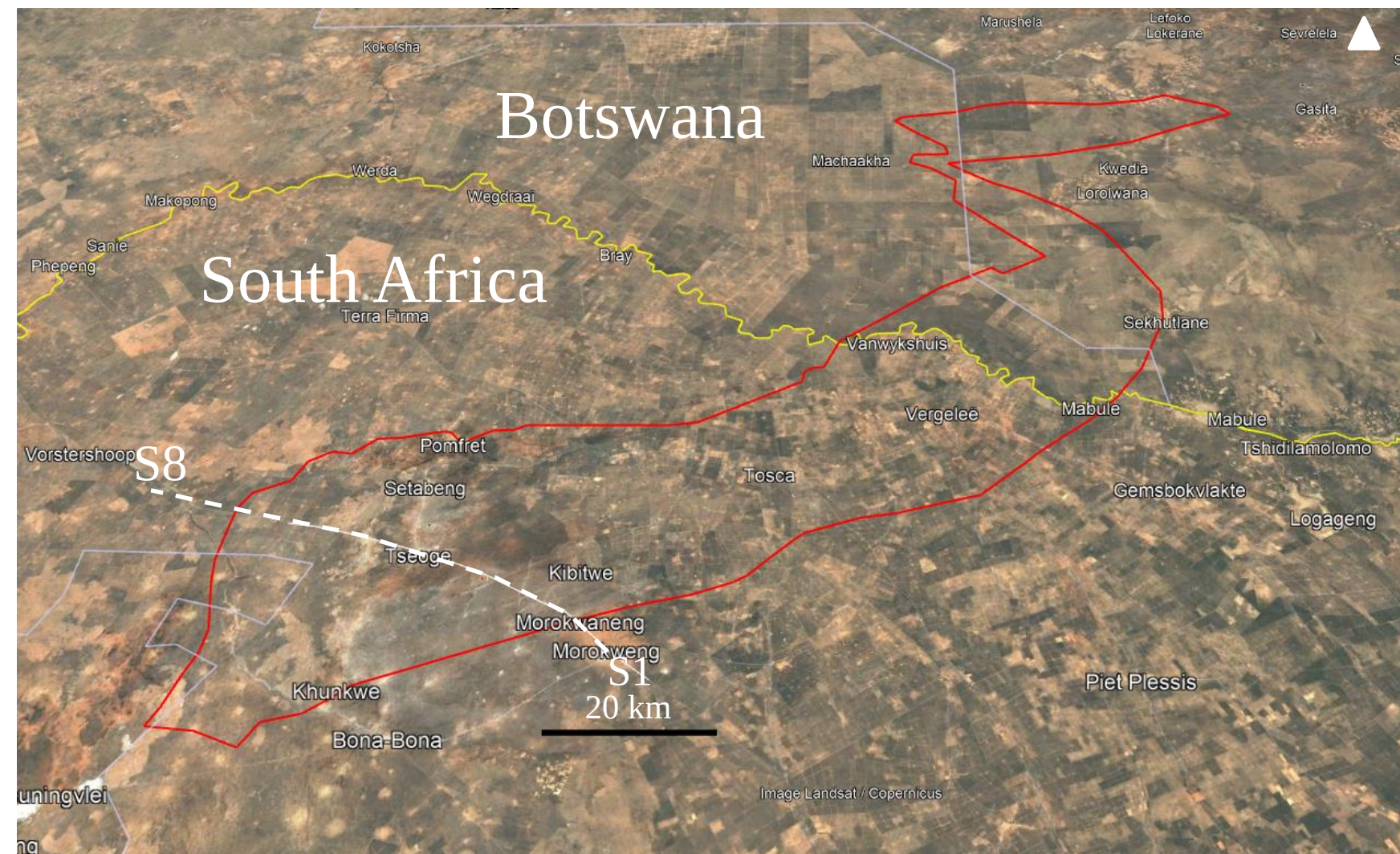
Weathered dolomite aquifer delineated from 170 to 210 d depth, EPD range: 0.042 to 0.083 mV



Examples of TEFSM application in groundwater exploration

Our field experience

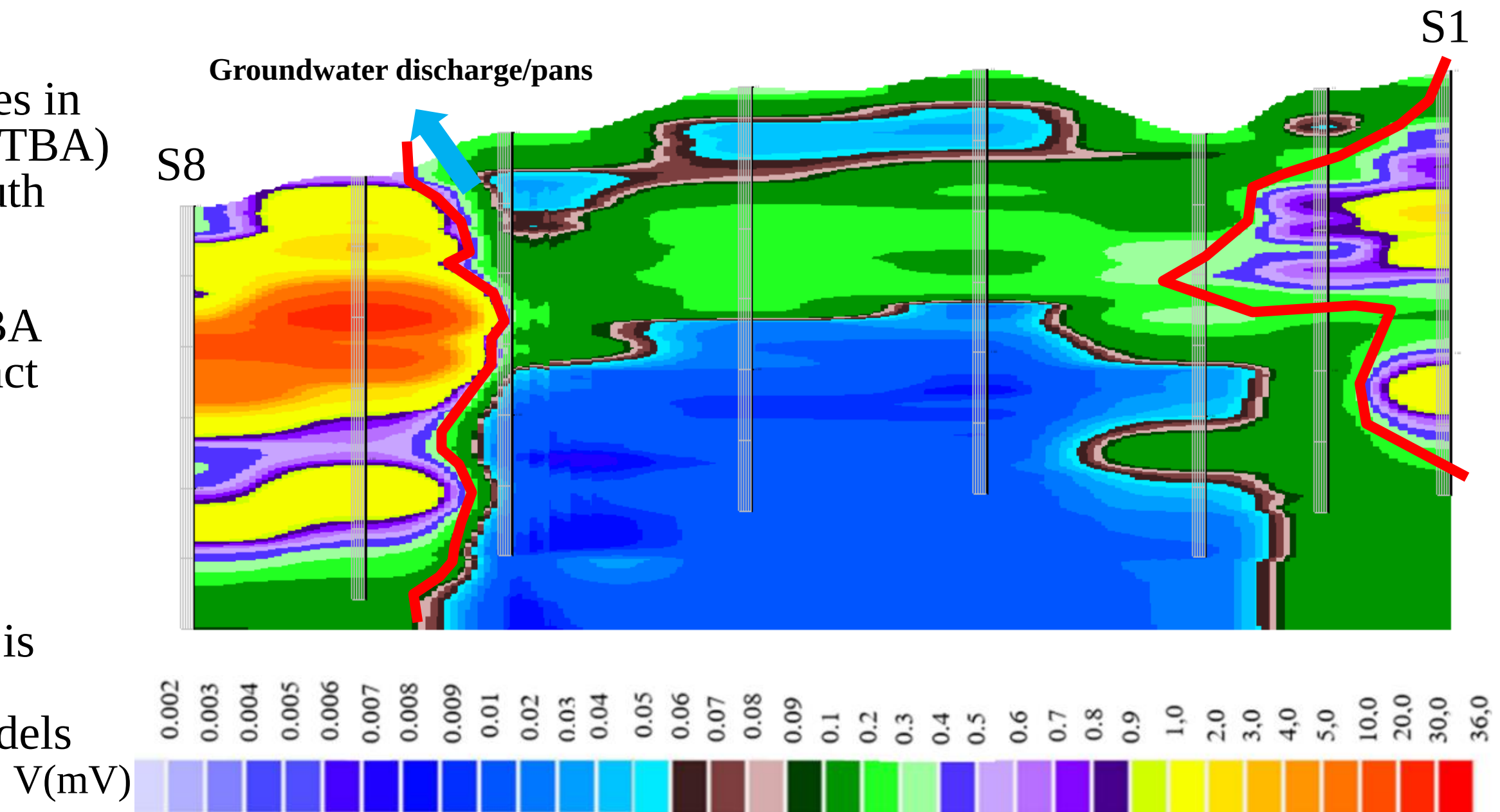
- Delineation of groundwater discharge zones in the Khakhea-Bray transboundary aquifer (TBA) shared by Botswana and South Africa South (Gomo and Ngobe 2024b)
- TEFSM corroborated with data from literature
- TEFSM 50 km survey line across the TBA (8 stations)



Examples of TEFSM application in groundwater exploration

Our field experience

- Delineation of groundwater discharge zones in the Khakhea-Bray transboundary aquifer (TBA) shared by Botswana and South Africa South (Ngobe 2022; Gomo and Ngobe 2024b)
- TEFSM delineated the boundary of the TBA where groundwater discharges at the contact zone into pans
- The pans at the discharge zone has water throughout the extended dry periods
- When borehole drilling and lithology data is available, the geophysical models can be converted to hydrogeology conceptual models for numerical modelling purposes



2 D geophysical model (Ngobe 2022)



What are we learning?

- The application of the approach is site-specific,
- The TEFSM measurements needs to be calibrated with drilling data/area hydrogeology from the site of interest before use,
- The approach does not detect water, it is however able to delineate vertical and lateral variations of lithology associated with discontinuities, weathered zones, and hard rocks, these can become potential targets for groundwater,
- Weathered zones or weaker formation appears as low EPD contrast
- The measurements appear to be influenced by the time of the day, seasons, and or weather patterns



What are we learning?

TEFSM capabilities

- Fast and real time measurements
- Low cost technology
- Deeper investigation
- Portable and compactible



What are we learning?

TEFSM challenges

The TEFSM is not magic, there are several limitations

- Lack of information about the principles of the approach, can raise questions and doubt
- Absence of standard data analysis guidelines lead to ineffective use of the approach but sometimes confusion
- There is no information about the impact of noise on the TEFSM, making data quality control difficult

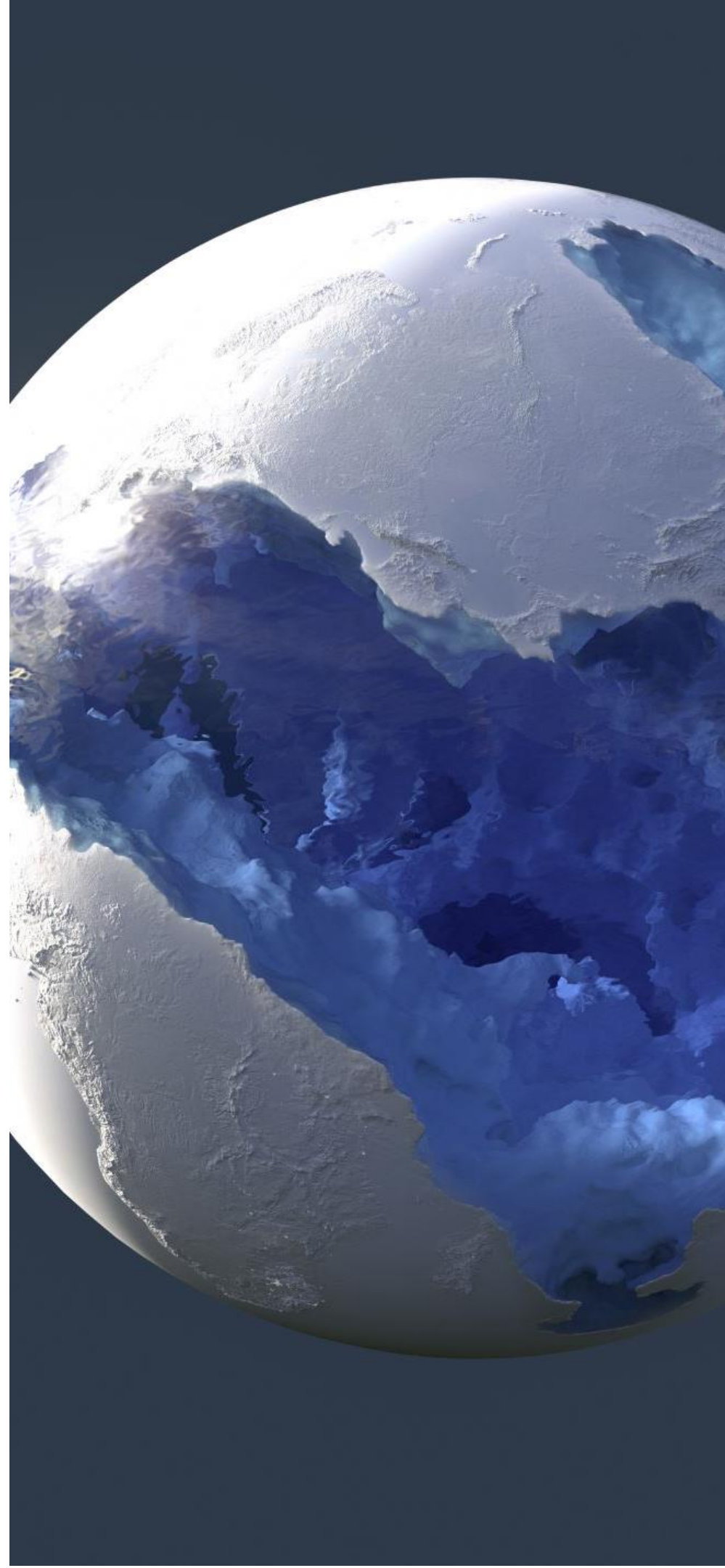


What are we learning?

TEFSM challenges

The TEFSM is not magic, there are several limitations

- The TEFSM does not measure an inherent property of rocks, hence it is not possible to define the standard electric potential difference associated with aquifers
- TEFSM readings are likely to vary from place because the natural EM field also varies, but when corroboration with existing site drilling data improves the interpretation (**learn as we go**)
- Approach can also detect aquitards as aquifers, but not useful for production purposes



Conclusions

- The TEFSM approach has great potential and been favorably received in many low-income regions
- Despite this potential, several limitations are also emerging,
- More research is required to improve the understanding of TEFSM application
- We cannot ignore the technology because many of our people are already using it unfortunately with limited scientific guidance due to the issues raised.

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